Axioms

1. *"A is above C, D is above F and on E."*
   $\phi_1 : Above(A, C) \wedge Above(E, F) \wedge On(D, E)$

2. *"A is green while C is not."*
   $\phi_2 : Green(A) \wedge \neg Green(C)$

3. *"Everything is on something."*
   $\phi_3 : \forall x \exists y. On(x, y)$

4. *"Everything that is free has nothing on it."*
   $\phi_4 : \forall x. (Free(x) \rightarrow \neg \exists y. On(y, x))$

5. *"Everything that is green is free."*
   $\phi_5 : \forall x. (Green(x) \rightarrow Free(x))$

6. *"There is something that is red and is not free."*
   $\phi_6 : \exists x. (Red(x) \wedge \neg Free(x))$

7. *"Everything that is not green and is above B, is red."*
   $\phi_7 : \forall x. (\neg Green(x) \wedge Above(x, B) \rightarrow Red(x))$

*

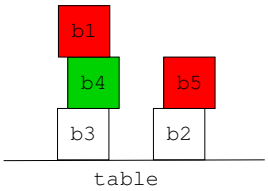
Exercise 3.13. ☼ ✍

Language *Constants:* A, B, C, D, E, F;
Predicates: $On^2, Above^2, Free^1, Red^1, Green^1$.

Interpretation *Let interpretation* $\mathcal{I}_1$ *be the following:*

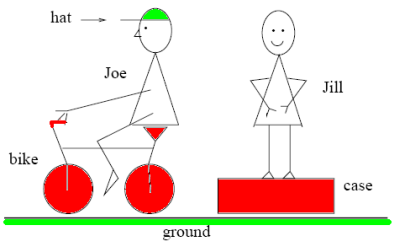
- $\mathcal{I}_1(A) = b_1$, $\mathcal{I}_1(B) = b_2$, $\mathcal{I}_1(C) = b_3$, $\mathcal{I}_1(D) = b_4$, $\mathcal{I}_1(E) = b_5$, $\mathcal{I}_1(F) = table$
- $\mathcal{I}_1(On) = \{\langle b_1, b_4\rangle, \langle b_4, b_3\rangle, \langle b_3, table\rangle, \langle b_5, b_2\rangle, \langle b_2, table\rangle\}$
- $\mathcal{I}_1(Above) = \{\langle b_1, b_4\rangle, \langle b_1, b_3\rangle, \langle b_1, table\rangle, \langle b_4, b_3\rangle, \langle b_4, table\rangle,$
  $\langle b_3, table\rangle, \langle b_5, b_2\rangle, \langle b_5, table\rangle, \langle b_2, table\rangle\}$

- $\mathcal{I}_1(Free) = \{\langle b_1\rangle, \langle b_5\rangle\}$, $\mathcal{I}_1(Green) = \{\langle b_4\rangle\}$, $\mathcal{I}_1(Red) = \{\langle b_1\rangle, \langle b_5\rangle\}$



And let interpretation $\mathcal{I}_2$ be:

- $\mathcal{I}_2(A) = hat$, $\mathcal{I}_2(B) = Joe$, $\mathcal{I}_2(C) = bike$, $\mathcal{I}_2(D) = Jill$, $\mathcal{I}_2(E) = case$, $\mathcal{I}_2(F) = ground$
- $\mathcal{I}_2(On) = \{\langle hat, Joe\rangle, \langle Joe, bike\rangle, \langle bike, ground\rangle, \langle Jill, case\rangle, \langle case, ground\rangle\}$
- $\mathcal{I}_2(Above) = \{\langle hat, Joe\rangle, \langle hat, bike\rangle, \langle hat, ground\rangle, \langle Joe, bike\rangle, \langle Joe, ground\rangle, \langle bike, ground\rangle, \langle Jill, case\rangle, \langle Jill, ground\rangle, \langle case, ground\rangle\}$
- $\mathcal{I}_2(Free) = \{\langle hat\rangle, \langle Jill\rangle\}$, $\mathcal{I}_2(Green) = \{\langle hat\rangle, \langle ground\rangle\}$, $\mathcal{I}_2(Red) = \{\langle bike\rangle, \langle case\rangle\}$



For each formula in Exercise 3.12, decide whether it is satisfied by $\mathcal{I}_1$ and/or $\mathcal{I}_2$.

Solution.

- $\mathcal{I}_1 \models \neg\phi_1 \wedge \neg\phi_2 \wedge \neg\phi_3 \wedge \phi_4 \wedge \neg\phi_5 \wedge \neg\phi_6 \wedge \phi_7$
- $\mathcal{I}_2 \models \phi_1 \wedge \phi_2 \wedge \neg\phi_3 \wedge \phi_4 \wedge \neg\phi_5 \wedge \phi_6 \wedge \phi_7$

*

Exercise 3.14.

Consider the following sentences:

1. *All actors and journalists invited to the party are late.*
2. *There is at least a person who is on time.*
3. *There is at least an invited person who is neither a journalist nor an actor.*

Formalize the sentences and prove that 3. is not a logical consequence of 1. and 2.

Solution.

1. $\forall x.((a(x) \vee j(x)) \wedge i(x) \rightarrow l(x))$
2. $\exists x.\neg l(x)$
3. $\exists x.(i(x) \wedge \neg a(x) \wedge \neg j(x))$

It's sufficient to find an interpretation $\mathcal{I}$ for which the logical consequence does not hold:

	l(x)	*a(x)*	*j(x)*	*i(x)*
Bob	*F*	*T*	*F*	*F*
Tom	*T*	*T*	*F*	*T*
Mary	*T*	*F*	*T*	*T*

*

Exercise 3.15.

Let $\Delta = \{1, 3, 5, 15\}$ and $\mathcal{I}$ be an interpretation on Δ interpreting the predicate symbols E^1 as 'being even', M^2 as 'being a multiple of' and L^2 as 'being less then', and s.t. $\mathcal{I}(a) = 1, \mathcal{I}(b) = 3, \mathcal{I}(c) = 5, \mathcal{I}(d) = 15$.
Determine whether $\mathcal{I}$ satisfies the following formulas:

1. $\exists y.E(y)$
2. $\forall x.\neg E(x)$
3. $\forall x.M(x,a)$
4. $\forall x.M(x,b)$
5. $\exists x.M(x,d)$
6. $\exists x.L(x,a)$
7. $\forall x.(E(x) \to M(x,a))$
8. $\forall x \exists y.L(x,y)$
9. $\forall x \exists y.M(x,y)$
10. $\forall x.(M(x,b) \to L(x,c))$
11. $\forall x \forall y.(L(x,y) \to \neg L(y,x))$
12. $\forall x.(M(x,c) \vee L(x,c))$

*

Exercise 3.16. 🍳 ✍

Provide a FOL language and a set of axioms that formalize the **graph coloring problem** *of a graph with at most n nodes, with connection degree $\leq m$, and with less then $k+1$ colors.*

- node degree*: number of adjacent nodes*
- connection degree *of a graph: max among all the degree of its nodes*
- graph coloring problem*: given a non-oriented graph, associate a color to each of its nodes in such a way that no pair of adjacent nodes have the same color.*

Solution.

Language

- *A unary function* color*, where* $\mathsf{color}(x)$ *is the color associated to the node* x
- *A unary predicate* node*, where* $\mathsf{node}(x)$ *means that* x *is a node*
- *A binary predicate* edge*, where* $\mathsf{edge}(x, y)$ *means that* x *is connected to* y

Axioms

1. $\forall x \forall y.(\mathsf{edge}(x, y) \rightarrow (\mathsf{color}(x) \neq \mathsf{color}(y))$
   "Two connected node are not equally colored."
2. $\forall x \forall x_1 \ldots \forall x_{k+1}. \left(\bigwedge_{h=1}^{k+1} \mathsf{edge}(x, x_h) \rightarrow \bigvee_{i,j=1,j\neq i}^{k+1} x_i = x_j\right)$
   "A node does not have more than k connected nodes."

*

Exercise 3.17. ✍

Let $\{c_1, .., c_k\}$ *be a non empty and finite set of colors. A partially colored directed graph is a structure* $\langle N, R, C\rangle$ *where*

- N *is a non empty set of nodes*
- R *is a binary relation on* N
- C *associates colors to nodes (not all the nodes are necessarily colored,. and each node has at most one color)*

Provide a first order language and a set of axioms that formalize partially colored graphs. Show that every model of this theory correspond to a partially colored graph, and vice-versa. For each of the following properties, write a formula which is true in all and only the graphs that satisfies the property:

1. *connected nodes don't have the same color*
2. *the graph contains only 2 yellow nodes*

3. *starting from a red node one can reach in at most 4 steps a green node*
4. *for each color there is at least a node with this color*
5. *the graph is composed of $|C|$ disjoint non empty subgraphs, one for each color*

Solution.

Language

- *a binary predicate* edge*, where* $\mathsf{edge}(n, m)$ *means that node n is connected to node m*
- *a binary predicate* color*, where* $\mathsf{color}(n, x)$ *means that node n has color x*
- *the following constants:* yellow, green, red

Axioms

0. "each node has at most one color"

$$\forall n \forall x.(\mathsf{color}(n, x) \rightarrow \neg \exists y.(y \neq x \wedge \mathsf{color}(n, y)))$$

1. "connected nodes do not have the same color"

$$\forall n \forall m \forall x.(\mathsf{edge}(n, m) \wedge \mathsf{color}(n, x) \rightarrow \neg \mathsf{color}(m, x))$$

2. "the graph contains only two yellow nodes"

$$\exists n \exists n'.(\mathsf{color}(n, \mathsf{yellow}) \wedge \mathsf{color}(n', \mathsf{yellow}) \wedge n \neq n' \wedge \\ \forall m.(m \neq n \wedge m \neq n' \rightarrow \neg \mathsf{color}(m, \mathsf{yellow})))$$

3. "starting from a red node one can reach in at most 4 steps a green node"

$$\forall n(\mathsf{color}(n, \mathsf{red}) \rightarrow \\ (\exists n_1.(\mathsf{edge}(n, n_1) \wedge \mathsf{color}(n_1, \mathsf{green})) \vee \\ \exists n_1, n_2.(\mathsf{edge}(n, n_1) \wedge \mathsf{edge}(n_1, n_2) \wedge \mathsf{color}(n_2, \mathsf{green})) \vee \\ \exists n_1, n_2, n_3.(\mathsf{edge}(n, n_1) \wedge \mathsf{edge}(n_1, n_2) \wedge \mathsf{edge}(n_2, n_3) \wedge \mathsf{color}(n_3, \mathsf{green})) \vee \\ \exists n_1, n_2, n_3, n_4.(\mathsf{edge}(n, n_1) \wedge \mathsf{edge}(n_1, n_2) \wedge \mathsf{edge}(n_2, n_3) \wedge \mathsf{edge}(n_3, n_4) \wedge \mathsf{color}(n_4, \mathsf{green})) \\ ))$$

4. "for each color there is at least a node with this color"

$$\forall x \exists n.\mathsf{color}(n, x)$$

5. "the graph is composed of $|C|$ disjoint non empty subgraphs, one for each color"

$$\forall x \exists n.\mathsf{color}(n, x) \wedge$$
$$\forall n \exists x.\mathsf{color}(n, x) \wedge$$
$$\forall n \forall x.(\mathsf{color}(n, x) \rightarrow \neg \exists y.(y \neq x \wedge \mathsf{color}(n, y))) \wedge$$
$$\forall n \forall m \forall x.(n \neq m \wedge \mathsf{color}(n, x) \wedge \mathsf{color}(m, x) \rightarrow$$
$$(\mathsf{edge}(n, m) \bigvee_{i=1}^{|N|} (\exists n_1, .., n_i.(\mathsf{edge}(n, n_1) \bigwedge_{j=1}^{i-1} \mathsf{edge}(x_j, x_j + 1) \wedge \mathsf{edge}(n_i, m)))))$$

*

Exercise 3.18. ✍

Minesweeper is a single-player computer game invented by Robert Donner in 1989. The object of the game is to clear a minefield without detonating a mine. The game screen consists of a rectangular field of squares. Each square can be cleared, or uncovered, by clicking on it. If a square that contains a mine is clicked, the game is over. If the square does not contain a mine, one of two things can happen: (1) A number between 1 and 8 appears indicating the amount of adjacent (including diagonally-adjacent) squares containing mines, or (2) no number appears; in which case there are no mines in the adjacent cells. An example of game situation is provided in the following figure:
Provide a first order language that allows to formalize the knowledge of a player in a game state. In such a language you should be able to formalize the following knowledge:

1. *there are exactly n mines in the minefield*
2. *if a cell contains the number 1, then there is exactly one mine in the adjacent cells.*

Figure 3.1: An example of a state in the Mines game

3. *show by means of deduction that there must be a mine in the position (3,3) of the game state of picture 3.1.*

Suggestion: define the predicate $Adj(x, y)$ to formalize the fact that two cells x and y are adjacent

Solution.

Language

1. *A unary predicate* mine*, where* $\mathsf{mine}(x)$ *means that the cell $x$contains a mine*

2. *A binary predicate* adj*, where* $\mathsf{adj}(x, y)$ *means that the cell x is adjacent to the cell y*

3. *A binary predicate* contains*, where* $\mathsf{contains}(x, n)$ *means that the cell x contains the number n*

Axioms

1. *There are exactly n mines in the game.*

$$\exists x_1, ..\exists x_n \left(\bigwedge_{i=1}^{n} \mathsf{mine}(x_i) \wedge \forall y \left(\mathsf{mine}(y) \rightarrow \bigvee_{i=1}^{n} y = x_i \right) \right)$$

2. *If a cell contains the number 1, then there is exactly one mine in the adjacent cells.*

   $\forall x.\,(\mathsf{contains}(x,1) \to \exists z.\,(\mathsf{adj}(x,z) \land \mathsf{mine}(z) \land \forall y.\,(\mathsf{adj}(x,y) \land \mathsf{mine}(y) \to y = z)))$

3. *Show by means of deduction that there must be a mine in the position (3,3)*
   from Picture 3.1 we have:

   a. $\mathsf{contains}((2,2),1)$

   b. $\neg\mathsf{mine}((1,1)) \land \neg\mathsf{mine}((1,2)) \land \neg\mathsf{mine}((1,3))$

   c. $\neg\mathsf{mine}((2,1)) \land \neg\mathsf{mine}((2,2)) \land \neg\mathsf{mine}((2,3))$

   d. $\neg\mathsf{mine}((3,1)) \land \neg\mathsf{mine}((3,2))$

   we can deduce:

   e. $\exists z.\,(\mathsf{adj}((2,2),z) \land \mathsf{mine}(z) \land \forall y.\,(\mathsf{adj}((2,2),y) \land \mathsf{mine}(y) \to y = z))$ $\rightsquigarrow$ *from a. and axiom 2*

   f. $\mathsf{mine}((1,1)) \lor \mathsf{mine}((1,2)) \lor \mathsf{mine}((1,3)) \lor \mathsf{mine}((2,1)) \lor \mathsf{mine}((2,2)) \lor \mathsf{mine}((2,3)) \lor \mathsf{mine}((3,1)) \lor \mathsf{mine}((3,2)) \lor \mathsf{mine}((3,3))$ $\rightsquigarrow$ *from e.*

   g. $\mathsf{mine}((3,3))$ $\rightsquigarrow$ *from b.,c.,d. and f.*

*

Exercise 3.19. ☁ ✍

Formalize in first order logic the train connections in Italy. Provide a language that allows to express the fact that a town is directly connected (no intermediate train stops) with another town, by a type of train (e.g., intercity, regional, interregional). Formalize the following facts by means of axioms:

1. *There is no direct connection from Rome to Trento*

2. *There is an intercity from Rome to Trento that stops in Firenze, Bologna and Verona.*

3. *Regional trains connect towns in the same region*

4. *Intercity trains don't stops in small towns.*

Solution. *We define the language as follows*

Constants $RM, FI, BO, VR, TN, \ldots$ *are identifiers of the towns of Roma, Firenze, Bologna, Verona, Trento, ...* and *InterCity, Regional, ...* are the identifiers of the type of trains

Predicates *Train with arity equal to 1, where Train*(x) *means* x *is a train*
Town with arity equal to 1, where Train(x) *means* x *is a town*
SmallTown with arity equal to 1, where Train(x) *means* x *is a small town*
TrainType with arity equal to 2, where TrainType(x, y) *means that the train* x *is of type* y.
IsInRegion with arity equal to 2, where IsInRegion(x, y) *means that the town* x *is in region* y. *DirectConn with arity equal to 3, where DirectConn*(x, y, z) *means that the train* x *directly connects (with no intermediated stops) the towns* y *and* z.

Background axioms *With these set of axioms we have to formalize some background knowledge which is necessary to make the formalization more adeguate*

1. *a train has exactly one train type;*

$$\forall x(\mathit{Train}(x) \rightarrow \exists y(\mathit{TrainType}(x, y))) \wedge \forall xyz(\mathit{TrainType}(x, y) \wedge \mathit{TrainType}(x, z) \rightarrow y = z) \tag{3.1}$$

2. *Intercity type is different from regional type:*

$$\neg(\mathit{InterCity} = \mathit{Regional}) \quad (\textit{also written as } \mathit{InterCity} \neq \mathit{Regional}) \tag{3.2}$$

3. *A town is associated to exactly one region*

$$\forall x(\mathit{Town}(x) \rightarrow \exists y(\mathit{IsInRegion}(x, y))) \wedge \forall xyz(\mathit{IsInRegion}(x, y) \wedge \mathit{IsInRegion}(x, z) \rightarrow y = z) \tag{3.3}$$